Facebook Social Media Connectivity

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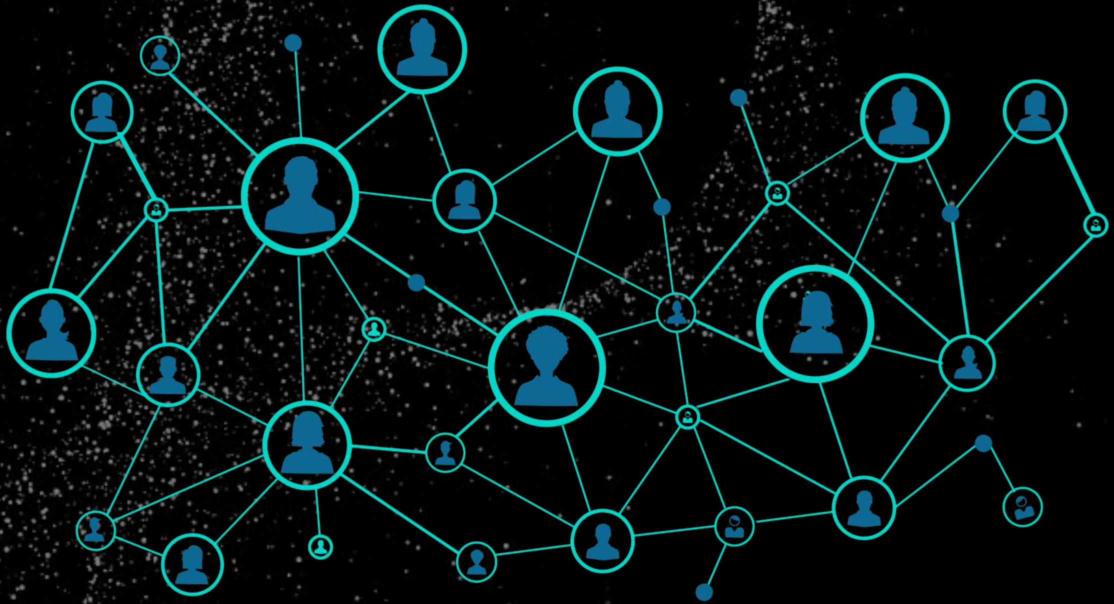
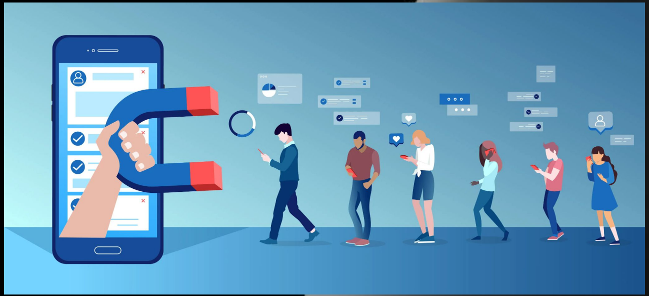


Business Scenario

Launch new campaign and want to maximize reach by targeting influential users.



Explore database technologies that can offer us advantages like more performance, higher scalability.

MongoDB

Redis

Neo4j

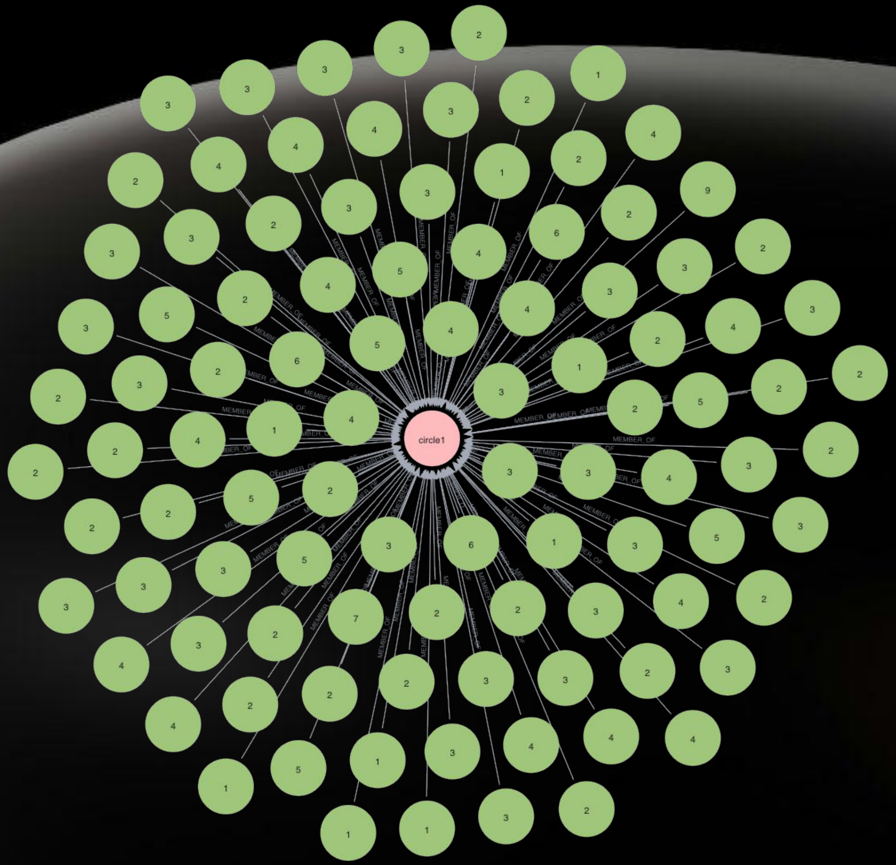
Graph Design

Goal
Model user relationships, community structures, and engagement behavior to support friend recommendations, community detection, and influence analysis.

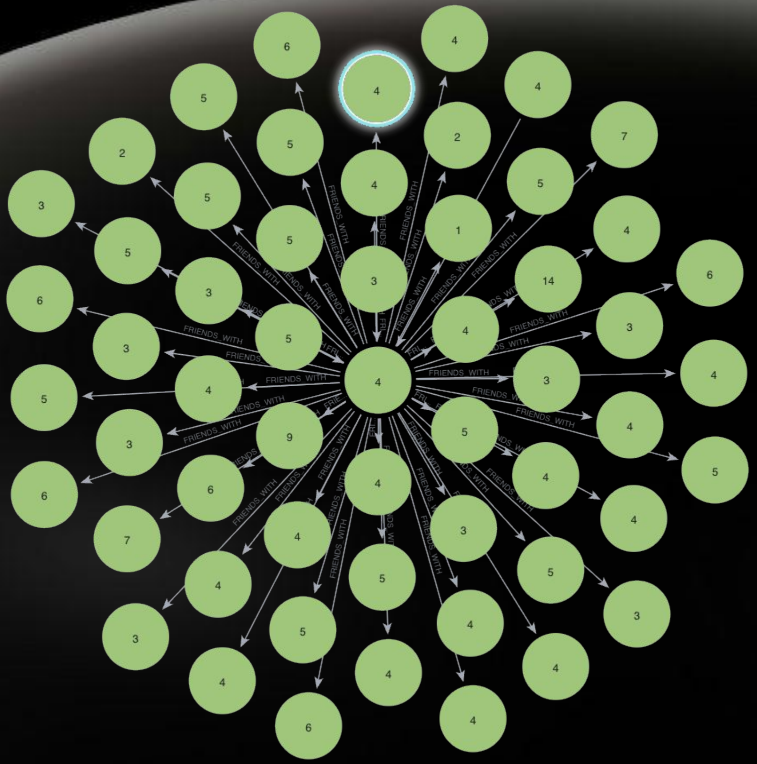
Nodes
User (user_id) → individual users in the network
Circle (circle_name, ego_id) → ground-truth friend groups (communities)
Post (post_id) → content created by users

Relationships
FRIENDS_WITH → connects users (undirected friendship network)
MEMBER_OF → connects users to circles (community membership)
POSTED → connects users to their posts
LIKED → connects users to posts they engage with

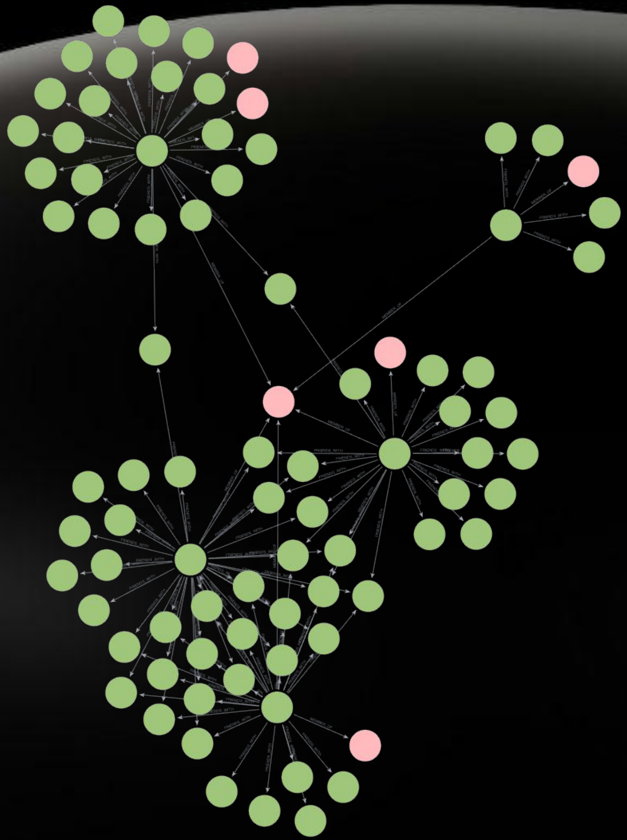
One Community "Circle"



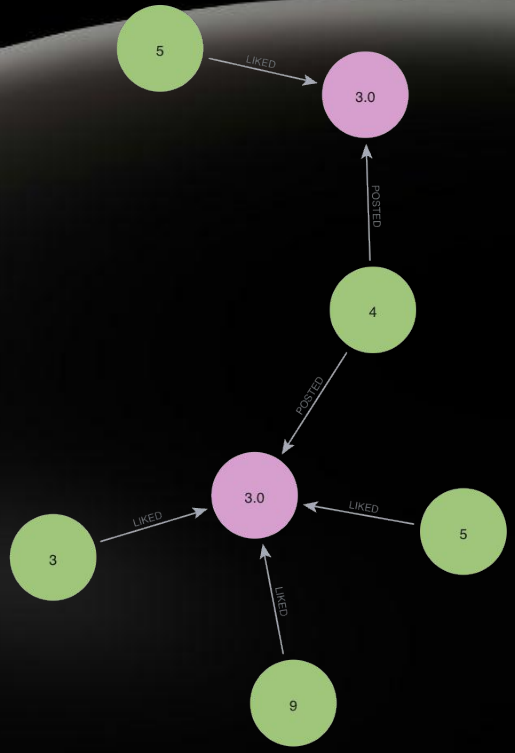
One User's Friendship Network



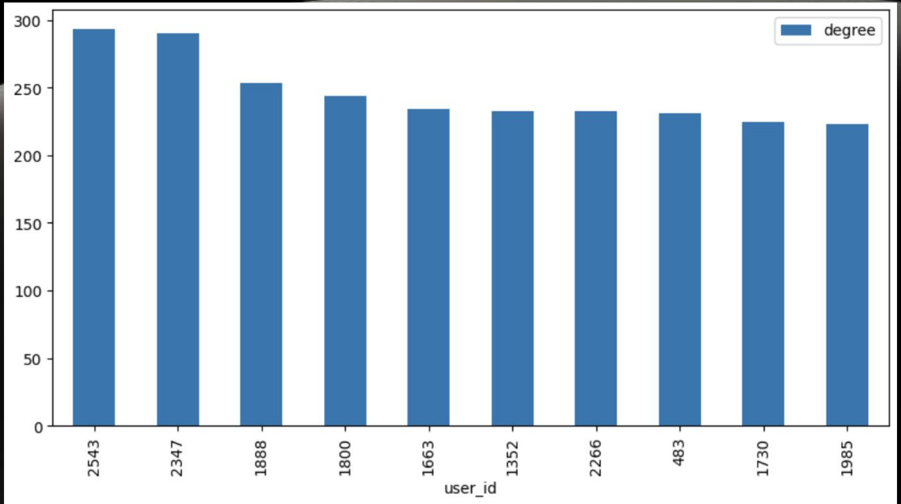
One Ego Network



One User's Posts and Who Liked



Degree Centrality Algorithm



Finds the most highly connected users in the network, allowing us to identify influential nodes that may play a key role in information dissemination and engagement.

Betweenness Centrality Algorithm

Degrees For Nodes 1085, 171

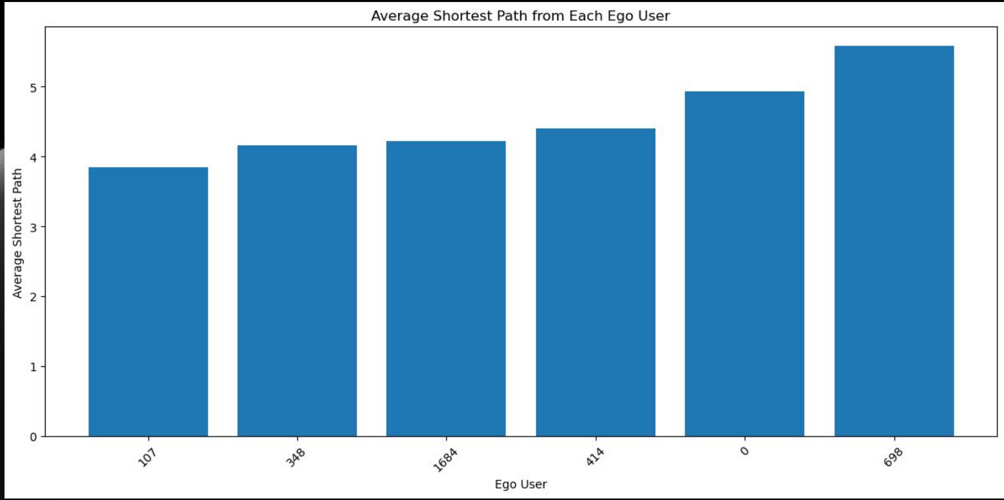
64

22

Graph with 4039 nodes and 84243 edges

	user_id	betweenness	is_ego
1085	1085	0.239393	0
1718	1718	0.162087	0
698	698	0.086914	1
1577	1577	0.086833	0
862	862	0.084959	0
1405	1405	0.082429	0
1165	1165	0.077259	0
136	136	0.069862	0
107	107	0.066364	1
171	171	0.063492	0

Shortest Path Algorithm



Finds the shortest connection between two users in the network by following friendship links. This graph shows the average shortest path length from each ego user to all other reachable users in the network. Lower values indicate that the ego user is more centrally positioned in the network.

Mongo DB Business case

Business case: MongoDB can be used to manage user-generated content in a social network platform such as; user profiles, posts, comments.

MongoDB is a good fit because it stores data in a document-based format, which makes it easier to handle semi-structured and changing data without constantly redesigning the schema.

A relational DB is not a good fit for this case because the data structure is highly variable and nested. The rigidity of relational db makes less efficient and less flexible than MongoDB in a fast changing platform..

Redis Business Case

Business case: Redis can be used to support fast access to frequently requested data in social network platform, such as; trending posts, popular users.

Redis can store these frequently accessed results in memory so they can be retrieved with very low latency. This is especially useful in a social platform where many users repeatedly request the same or similar high-traffic data.

A relational DB is not a good fit for this case because it is not optimized for fast in-memory access and high frequency caching workloads. If every feed request or ranking lookup had to be processed directly through a relational DB, the system could place unnecessary load and become slow.

Thank you